

Product Requirements Document: Ace-app

Project Version: 1.0 (Hackathon Build) **Core Tech:** React Native (Expo), Firebase (Firestore/Auth/Cloud Messaging), Expo-Location. **Philosophy:** "Discipline over Utility" — Physical presence is the only currency. The system enforces fairness through strict geofencing.

1. Product Concept

Ace-app is a location-locked court management system for tennis complexes (Hubs). It uses high-accuracy background geofencing to ensure that only players physically present at a facility can occupy the waitlist or maintain control of an active court.

2. User Authentication & Identity

2.1 Login/Signup

- **Requirements:** Unique Username, Email, and Password.
- **Identity:** Users must be authenticated to join a waitlist or claim a court.
- **Phone Number:** Optional field. Collected for "Critical Alert" SMS fallback if Push Notifications are disabled or data connectivity is lost.
- **Permissions:** Upon first login, users must grant "Always Allow" location permissions to enable background tracking (The Silent Sentry).

3. The "Hub" Architecture

3.1 Hub Definition

A "Hub" is a single fenced-in tennis complex containing multiple courts.

- **Geofence Logic:** A circular or polygonal perimeter drawn around the physical fence with a **10m buffer** to account for GPS drift.
- **Center Point:** Each Hub has a defined Latitude/Longitude.
- **Standard Radius:** 50 meters (default).

4. Operational Logic (The Rules of Play)

4.1 The "Tag Holder" (Group Leader)

- **Logic:** Only one user per group (the Tag Holder) is geotracked.

- **Responsibility:** If the Tag Holder's device leaves the geofence or dies, the entire group is subject to removal.
- **Input:** When joining, the Leader must specify N (players) and "Open to Join" (Yes/No).

4.2 The "5+5 Silent Sentry" (Active Court Management)

This logic applies to players currently assigned to a court.

- **Phase 1 (0–5 Mins):** User exits the geofence. The system monitors silently. No notifications.
- **Phase 2 (5–10 Mins):** At the 5:00 mark, a high-priority **Live Push Notification** is sent: *"You've left the Hub. Return within 5 minutes to keep your court."*
- **The Return:** If the GPS re-enters the geofence before the 10:00 mark, the timer resets to zero automatically.
- **The Release:** If the user is not back by 10:01, the court status flips to `Open`.

4.3 Waitlist & "Claim" Logic

- **Joining:** The "Join Waitlist" button is only enabled if the user's GPS is currently within the Hub geofence.
- **Waitlist Exit:** If a waitlisted user leaves the fence, they have **3 minutes** to respond to a "Stay on List?" prompt.
- **Claiming:** When a court becomes `Open`, the #1 user is pinged. They have **3 minutes** to tap "Claim."
- **Claim Verification:** The "Claim" button is only functional if the user is physically inside the geofence at the moment of tapping.

5. Technical Requirements

5.1 Firebase Firestore Schema

- `hubs/{hubId} : { name: string, center: geopoint, total_courts: number }`
- `courts/{courtId} : { hub_id: string, status: "active"|"pending"|"open", tag_holder_id: uid, expires_at: timestamp }`
- `waitlist/{userId} : { username: string, position: timestamp, group_size: number, is_open: boolean }`
- `users/{userId} : { username: string, fcmToken: string, last_inside: timestamp }`

5.2 Notification System

- Must use **Firebase Cloud Messaging (FCM)** for background notifications.
- Notifications must trigger even when the app is closed/backgrounded.

6. UX/UI Requirements

- **Dashboard:** Real-time 2-column grid showing court statuses.
- **Status Colors:** * **Green:** Open/Available.
 - **Clay/Red:** Active/Occupied.
 - **Pulsing Orange:** Pending Vacancy (Show countdown timer).
- **The "Couch Mode" (Debug):** A developer toggle to simulate "Exiting Geofence" without physical movement.

7. Global Configuration Constants

```
const CONFIG = {  
  GEOLOCK_RADIUS: 50,           // Meters  
  WAITLIST_EXIT_GRACE: 180,     // 3 Minutes  
  SENTRY_PHASE_1: 300,         // 5 Minutes (Silent)  
  SENTRY_PHASE_2: 300,         // 5 Minutes (Warning)  
  CLAIM_WINDOW: 180,           // 3 Minutes  
  GPS_POLL_INTERVAL: 20000     // 20 Seconds  
};
```